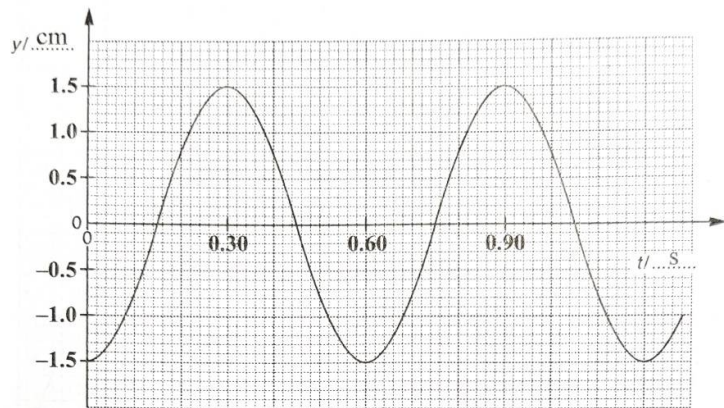


4(a)



4(b)(i)

$$\text{Period} = \frac{12.0}{20} = 0.60 \text{ s}$$

Equation of graph is

$$y = -15 \cos\left(\frac{2\pi}{T}t\right)$$

$$= -15 \cos\left(\frac{2\pi}{0.6}t\right)$$

Displacement 2.0 mm:

$$2.0 = -15 \cos\left(\frac{2\pi}{0.6}t\right)$$

$t_1 = 0.1628$ s (displacement 2.00 mm, on the way down)

Displacement -2.0 mm:

$$-2.0 = -15 \cos\left(\frac{2\pi}{0.6}t\right)$$

$t_1 = 0.1372$ s (displacement 2.00 mm, on the way down)

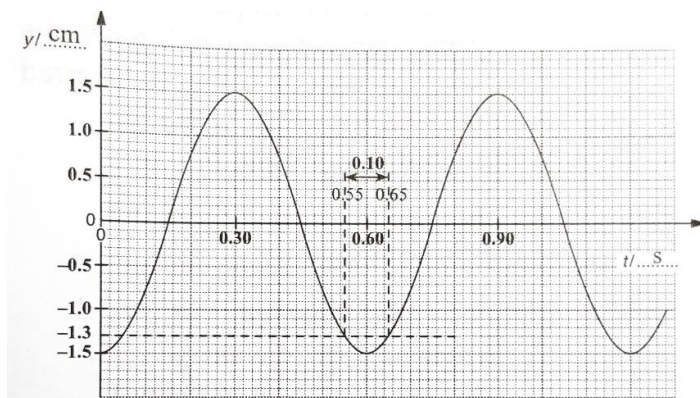
Minimum time taken

$$= 0.1628 - 0.1372$$

$$= 0.0256$$

$$\approx 0.026$$
 s

4(b)(ii)



From Fig. 4.2, the time interval = $0.65 - 0.55 = 0.10$ s.

4(c)

The timing should be made from the position where the displacement is zero, i.e. equilibrium position. The pendulum makes large observable displacements near its equilibrium position, e.g. the displacement of -0.5 mm to the displacement of +0.5 mm takes less than 0.10 s. From results in (b)(ii), the pendulum's small displacements near its maximum displacement make it difficult to observe if it has truly reached its maximum position, easily causing an error in recorded period of more than 0.10 s.

5(a)(i)

The first minima are located at angular displacements of θ